CineRec: A Practical Multi-Modal Recommender Study with Cross-Dataset Validation and Published-Baseline Comparison

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ABSTRACT

Recommender systems have evolved from simple neighborhood-based methods to complex deep learning models that incorporate side information such as text and images. However, many comparative studies focus on a single dataset and lack rigorous statistical validation against published baselines. We present CineRec, a practical study that evaluates six recommendation models of increasing complexity—UserCF, ItemCF, SVD, NeuMF, LightGCN, and MultiModalNCF—on two domains: MovieLens 100K and Amazon Video_Games. We reproduce two published baselines (BPR-MF and VBPR) under the same evaluation protocol, report results as mean±standard deviation over five random seeds, and conduct both modality ablation and cold-start studies. Our findings show that (1) the ranking of models is largely consistent across domains, (2) content features provide measurable gains in cold-start settings but have limited impact on overall ranking quality, and (3) MultiModalNCF performs on par with VBPR, a well-established content-aware baseline. We release the full codebase and evaluation scripts, and provide instructions to reproduce all datasets from public sources, to support reproducibility.

CCS CONCEPTS

• **Information systems** → **Recommender systems**.

KEYWORDS

Recommender systems, multi-modal recommendation, cold-start, cross-dataset validation, BPR-MF, VBPR

1 INTRODUCTION

Recommender systems are a core component of modern information retrieval, powering personalized content discovery across e-commerce, media, and social platforms. The field has seen a rapid evolution of methods, from early collaborative filtering approaches based on user-item interaction matrices to modern deep learning models that incorporate rich side information such as text descriptions, images, and knowledge graphs.

Despite this rapid progress, many comparative studies in recommender systems suffer from three limitations: (1) they are evaluated on a single dataset, limiting the generalizability of conclusions, (2) they lack comparison against published baselines under matched conditions, and (3) results are often reported as single-point estimates without statistical significance testing.

In this work, we present CineRec, a practical study designed to address these gaps. Our contributions are as follows:

(1) We implement and evaluate six recommendation models spanning the full evolution from classical neighborhood methods to deep multi-modal models, under a unified evaluation protocol.
(2) We reproduce two published baselines—BPR-MF [10] and VBPR [4]—on the same split, allowing direct comparison against established methods.
(3) We validate all conclusions across two datasets: MovieLens 100K and Amazon Video_Games, testing whether observed patterns generalize across domains.
(4) We report all neural model results as mean±standard deviation over five random seeds, distinguishing real effects from seed noise.
(5) We conduct both modality ablation and cold-start studies, quantifying the contribution of each content channel.

The study design is summarised in Figure 1. The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes our methodology and model suite. Section 4 presents experimental setup and results. Section 5 discusses findings and limitations. Section 6 concludes.

2 RELATED WORK

2.1 Neighborhood-Based Collaborative Filtering

User-based and item-based collaborative filtering [1] are among the earliest and simplest recommendation approaches. They rely on similarity measures between users or items to generate recommendations. While simple, they remain competitive baselines and serve as the foundation for more complex methods.

2.2 Matrix Factorization

Matrix factorization methods such as SVD [8] and BPR [10] transform the recommendation problem into a latent factor space. BPR, in particular, introduced a pairwise ranking loss that has become a standard for implicit feedback recommendation.

2.3 Neural Collaborative Filtering

NeuMF [6] combined matrix factorization with multi-layer perceptrons to model non-linear user-item interactions, demonstrating that deep learning can improve recommendation quality over traditional matrix factorization.

2.4 Graph-Based Recommendation

LightGCN [5] simplified graph convolutional networks for recommendation by removing feature transformation and non-linear activation, achieving strong performance with increased efficiency.

Independent research preprint; not submitted for publication.

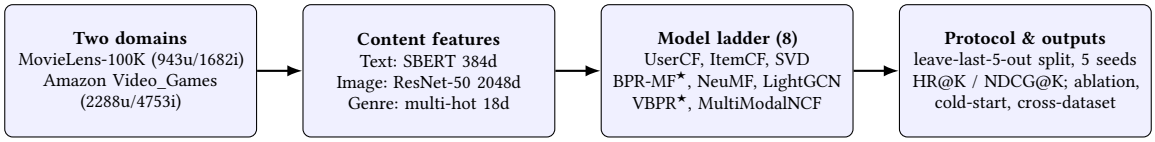


Figure 1: Overview of the CineRec study. Two domains feed precomputed text/image/genre content features into a ladder of eight recommenders—six ladder models plus two reproduced published baselines (marked ★)—all evaluated under a single leave-last-5-out protocol with five-seed mean±std reporting.

2.5 Multi-Modal Recommendation

VBPR [4] extended BPR with visual features, demonstrating that content information can improve recommendation accuracy, especially for cold-start items. Multi-modal recommendation has since become an active research area, incorporating text, images, and other side information.

3 METHODOLOGY

3.1 Model Suite

We evaluate six models of increasing complexity, plus two published baselines. Table 1 summarizes all model configurations; Figure 2 details the MultiModalNCF architecture.

Key details:

- **NeuMF**: Neural Matrix Factorization combining GMF and MLP branches.
- **LightGCN**: Simplified graph convolutional network with BPR loss.
- **MultiModalNCF**: Our multi-modal extension of NeuMF. A content tower projects each modality (text 384d, image 2048d, genre 18d) to 32 dimensions each, concatenates to 96d, then projects to 32d before fusing with the behavioural embedding.
- **Positive threshold**: Neural models binarise ratings ≥ 4 as positive during training.
- **Negative sampling**: 4 negatives per positive for NeuMF and MultiModalNCF; BPR-based models use standard BPR sampling.

3.2 Evaluation Protocol

We use the standard leave-last-5-out split: the five most recent interactions of each user are held out as the test set. All models are evaluated against the same candidate set (all items minus the user's training interactions). We report HR@K and NDCG@K at $K \in \{10, 20\}$; Recall@10 is additionally plotted in Figure 4.

Neural models and both baselines are trained with five random seeds (42–46) and reported as mean±standard deviation. Deterministic models (UserCF, ItemCF, SVD) have zero spread. Figure 3 summarises the protocol.

3.3 Content Features

We precompute content features shared across all content-aware models:

- **Text**: Sentence-BERT (all-MiniLM-L6-v2) [9] embeddings of movie overviews, 384-dimensional.
- **Image**: ResNet-50 (pool5) [3] embeddings of movie posters, 2048-dimensional.
- **Genre**: Multi-hot genre vectors, 18-dimensional.

4 EXPERIMENTS

All experimental results below are read from the committed result artefacts of release v1.0.0 (2026-10-13) and are reproducible via the repository's make targets; the CI figures are as reported by the CI run on v1.0.0.

4.1 Datasets

We conduct experiments on two datasets:

- **MovieLens 100K** [2]: 943 users, 1,682 items, 100,000 ratings (1–5 scale). Users with ≥ 20 ratings.
- **Amazon Video_Games** [7]: 2,288 users, 4,753 items, 75,980 ratings (1–5 scale). Heavy users with ≥ 20 ratings.

The Amazon slice is built deterministically from the 5-core Amazon Reviews 2023 Video_Games dataset by keeping users with ≥ 20 ratings and re-applying the item 5-core rule. The original 5-core dataset is publicly available from the Amazon Reviews dataset repository; our slice definition and processing scripts are included in the codebase.

Poster and overview metadata are retrieved from the TMDB API and remain subject to TMDB's terms; MovieLens 100K is used under its non-commercial research terms, and the project's code is released under the MIT licence.

4.2 Main Results

Table 2 shows the main model comparison on MovieLens 100K. Key observations:

- The expected ordering UserCF < ItemCF < SVD holds, with the neural band (0.285–0.371 HR@10) sitting above SVD.
- BPR-MF and NeuMF are within one combined standard deviation of each other.
- VBPR's HR@10 edge over MultiModalNCF (0.0055) is smaller than one MultiModalNCF standard deviation, while MultiModalNCF leads both NDCG columns—the pair is read as statistically tied.

Figure 4 visualises the same comparison at K=10.

4.3 Modality Ablation

Table 3 shows the modality ablation study on MultiModalNCF. Removing all content lowers every metric, but by roughly one seed standard deviation. The single-modality deltas sit inside the noise

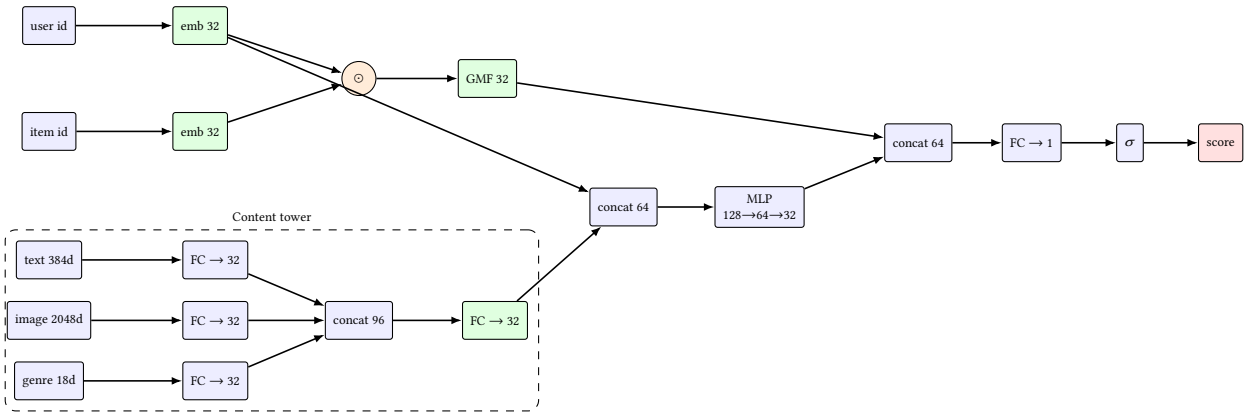


Figure 2: MultiModalNCF architecture (the configuration used in all reported runs). Behavioural path: GMF over user/item embeddings (d=32). Content tower: text/image/genre are each projected to 32 dims, concatenated to 96, then projected to a 32-dim content embedding; the MLP path replaces the item embedding with this content embedding. The two paths are concatenated (64) and mapped to a score. In the cold-start path the GMF item vector is zeroed, so only the content tower drives scoring.

Table 1: Model configurations

Model	Configuration
UserCF	Pearson similarity, $k = 50$ neighbors
ItemCF	Adjusted cosine similarity, $k = 50$ neighbors
SVD	Truncated SVD, $k = 64$ latent dimensions
BPR-MF	dim 64, epochs 50, lr 0.001, batch 4096, patience 8
NeuMF	dim 32, MLP (128, 64, 32), epochs 10, batch 512, 4 negatives, patience 3
LightGCN	dim 64, 3 layers, epochs 50, batch 4096, BPR objective, patience 8
VBPR	behaviour dim 64, visual dim 64, epochs 30, lr 0.005
MultiModalNCF	dim 32, MLP (128, 64, 32), epochs 10, batch 512; content tower projects 384/2048/18→32 each, concatenated (96)→32

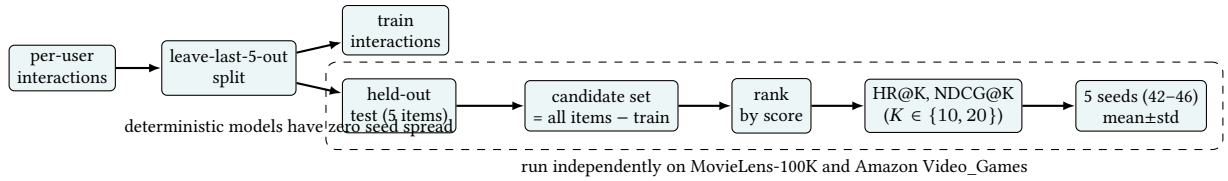


Figure 3: Evaluation protocol. A per-user leave-last-5-out split defines training and held-out interactions; held-out items are ranked against a candidate set of all non-training items; HR@K and NDCG@K ($K \in \{10, 20\}$) are reported over five seeds (42–46) as mean±std. The protocol is run independently on both domains.

(one even reverses sign), so no per-modality ranking is claimed. Figure 5 plots the per-variant HR@10 / NDCG@10.

4.4 Cold-Start Study

Table 4 shows the cold-start experiment, where 100 items are removed from training and evaluated as cold-start items. The ordering content > blind > random holds on every metric. The HR@10 spread of the content variant (0.0290) exceeds the gap to NeuMF (0.0193), so the margin is reported as a direction rather than a point magnitude. Figure 6 visualises the three-way comparison.

4.5 Cross-Dataset Validation

Table 5 compares performance across both datasets, and Table 6 repeats the modality ablation on Amazon.

What transfers: the ladder ordering, the BPR-MF ≈ NeuMF tie, the "single-modality deltas are inside seed noise" conclusion (on Amazon removing any single modality moves HR@10 by at most 0.0030, still inside the 0.0085 full-model spread), and the cold-start direction—the content model reaches HR@10 0.2364±0.0202 against 0.1204±0.0120 for the content-blind model and 0.1092 for random, a content/blind ratio of 1.96× (1.17× on MovieLens). What does not: on Amazon the top HR@10 point estimate belongs to LightGCN

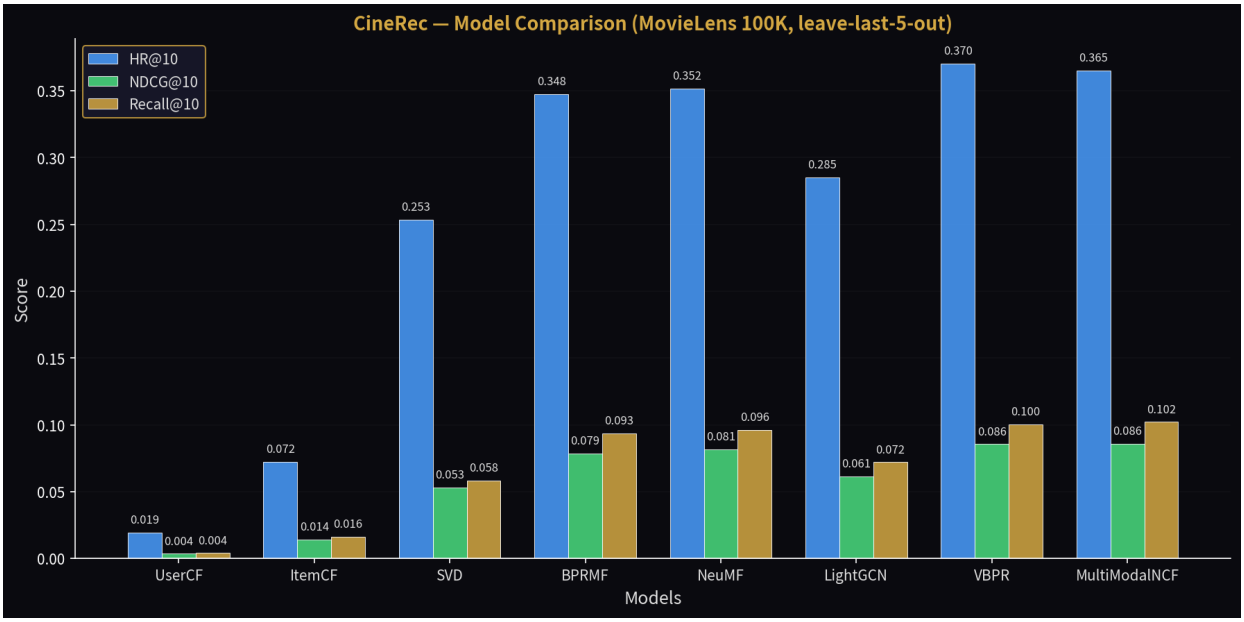


Figure 4: Model comparison on MovieLens 100K at K=10 (HR, NDCG, Recall). HR@10 and NDCG@10 bars are the 5-seed means from Table 2; Recall@10 is from the same evaluation artefacts.

Table 2: Model comparison on MovieLens 100K (mean±std over 5 seeds; train time in seconds)

Model	HR@10	NDCG@10	HR@20	NDCG@20	Train time (s)
UserCF	0.0191±0.0000	0.0035±0.0000	0.0477±0.0000	0.0060±0.0000	0.1
ItemCF	0.0721±0.0000	0.0142±0.0000	0.1283±0.0000	0.0199±0.0000	0.3
SVD	0.2534±0.0000	0.0532±0.0000	0.3542±0.0000	0.0664±0.0000	0.3
BPR-MF	0.3476±0.0116	0.0785±0.0018	0.4831±0.0116	0.1034±0.0018	10.6
NeuMF	0.3516±0.0036	0.0814±0.0016	0.4978±0.0073	0.1083±0.0017	31.4
LightGCN	0.2850±0.0041	0.0614±0.0004	0.4250±0.0047	0.0816±0.0007	10.7
VBPR	0.3705±0.0048	0.0855±0.0009	0.5200±0.0096	0.1130±0.0008	41.9
MultiModalNCF	0.3650±0.0077	0.0859±0.0017	0.5073±0.0148	0.1144±0.0015	73.2

Table 3: Modality ablation on MultiModalNCF (MovieLens 100K)

Variant	HR@10	NDCG@10	HR@20	NDCG@20
Full	0.3650±0.0077	0.0859±0.0017	0.5073±0.0148	0.1144±0.0015
w/o Text	0.3642±0.0100	0.0855±0.0024	0.5075±0.0108	0.1144±0.0025
w/o Image	0.3631±0.0085	0.0842±0.0026	0.5080±0.0029	0.1132±0.0025
w/o Genre	0.3673±0.0080	0.0857±0.0015	0.5107±0.0139	0.1144±0.0016
Behaviour only	0.3565±0.0061	0.0844±0.0024	0.4931±0.0089	0.1123±0.0033

(0.1187 vs 0.1164), a difference well inside the combined spread, and Behaviour-only ≈ Full (0.1163 vs 0.1164).

4.6 Reproducibility and Engineering Quality

To support reproducibility, we maintain a well-tested codebase with continuous integration:

- **Test suite**: 125 tests passed, with 50.9% line coverage (minimum threshold 45%). Tests cover data loading, model training loops, and evaluation metrics.
- **CI pipeline**: Three jobs run on every push to main and every pull request: linting + tests with coverage, end-to-end accessibility tests, and Docker image builds for both deployment modes (lite and full).

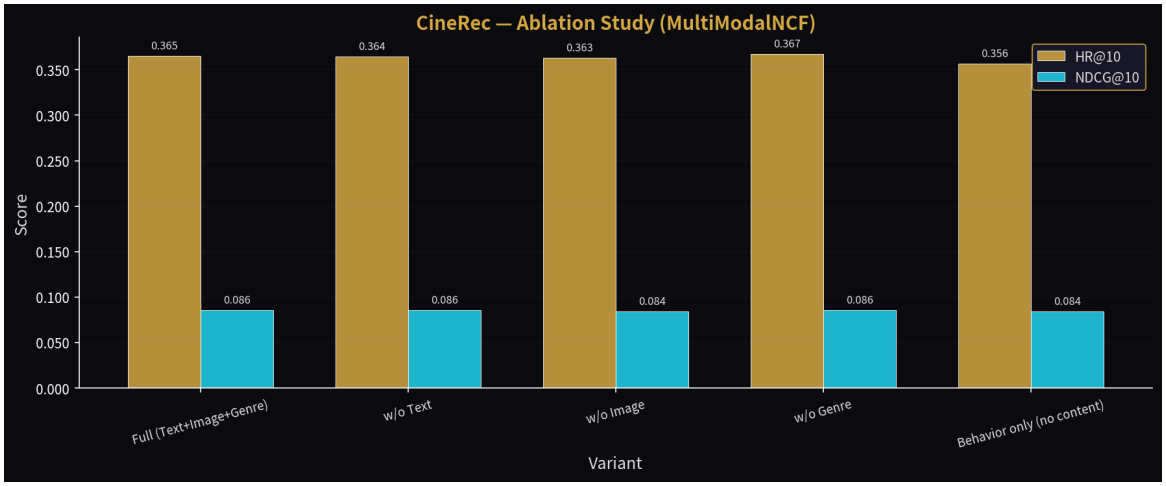


Figure 5: Modality ablation on MultiModalNCF (MovieLens 100K): HR@10 and NDCG@10 for the full model and each ablated variant. Values match Table 3.

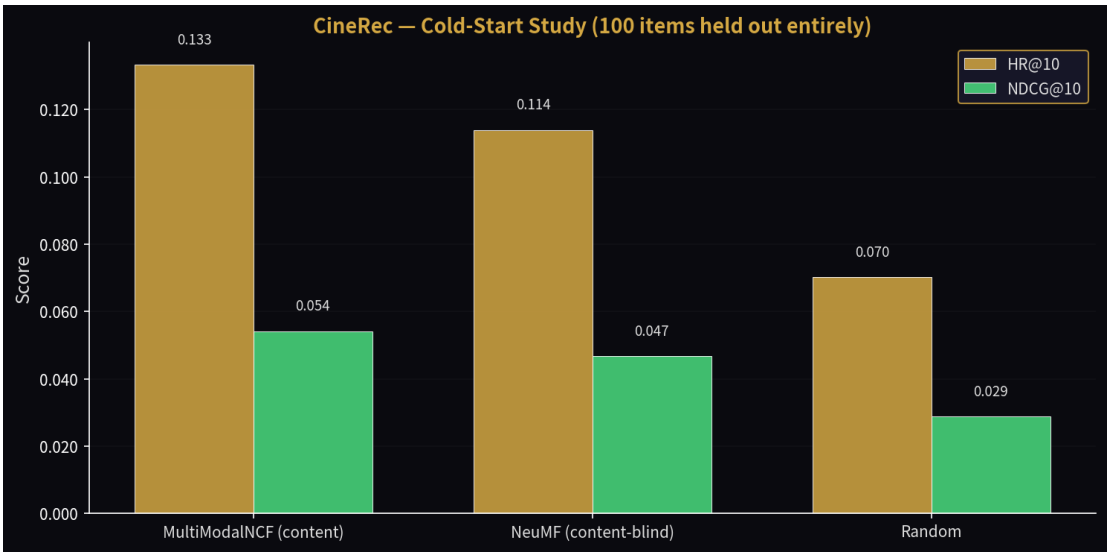


Figure 6: Cold-start study: 100 items held out of training and serving as the only candidates, comparing a content-aware model, a content-blind model and a random ranker. Values match Table 4.

Table 4: Cold-start performance (100 items removed from training)

Model	HR@10	NDCG@10	HR@20	NDCG@20
MultiModalNCF (content)	0.1333±0.0290	0.0542±0.0151	0.2754±0.0233	0.0882±0.0139
NeuMF (content-blind)	0.1140±0.0147	0.0467±0.0069	0.2351±0.0129	0.0739±0.0038
Random	0.0702	0.0288	0.2018	0.0610

- **Dependency hygiene**: We use pinned version lower bounds to ensure reproducible environments. Initial vulnerability scans flagged known advisories in transitive dependencies; after updating to patched versions, no known vulnerabilities remain (as of October 2026).
- **Documentation**: Key architectural decisions are documented in Architecture Decision Records (ADRs). The codebase includes make commands for setup, data preparation, training, and evaluation, allowing full reproduction from scratch.

Table 5: Cross-dataset validation: HR@10 / NDCG@10 comparison

Model	ML-100K HR@10	Amazon HR@10	ML-100K NDCG@10	Amazon NDCG@10
UserCF	0.0191	0.0122	0.0035	0.0017
ItemCF	0.0721	0.0214	0.0142	0.0035
SVD	0.2534	0.0651	0.0532	0.0116
BPR-MF	0.3476	0.1055	0.0785	0.0198
NeuMF	0.3516	0.1053	0.0814	0.0190
LightGCN	0.2850	0.1187	0.0614	0.0222
VBPR	0.3705	0.1097	0.0855	0.0196
MultiModalNCF	0.3650	0.1164	0.0859	0.0213

Table 6: Modality ablation on Amazon Video_Games (MultiModalNCF, HR@10, mean±std over 5 seeds)

Variant	HR@10
Full	0.1164±0.0085
w/o Text	0.1159±0.0050
w/o Image	0.1177±0.0033
w/o Genre	0.1194±0.0017
Behaviour only	0.1163±0.0011

- **Repository hygiene**: Pre-commit hooks reject large files, malformed YAML, and merge-conflict markers before each commit, and the tracked text files contain no credentials or secret material.

Figure 7 reports the per-model training cost behind Table 2.

5 DISCUSSION AND LIMITATIONS

5.1 Key Findings

(1) **Model ordering is largely consistent across domains.** The ladder backbone (UserCF < ItemCF < SVD) and the neural band's lead over SVD hold on both MovieLens and Amazon, suggesting these patterns are not dataset-specific artifacts. Which model leads *within* the neural band is domain-dependent, however (VBPR/MultiModalNCF on MovieLens, LightGCN on Amazon), so no cross-domain "best model" is claimed.

(2) **Content features help in cold-start, but not much in overall ranking.** Modality ablation shows that removing content features has limited impact on overall ranking quality, but cold-start experiments show measurable gains when content is available.

(3) **MultiModalNCF is competitive with established baselines.** Despite using a simple concatenation-based fusion, MultiModalNCF performs on par with VBPR, a well-established content-aware baseline.

5.2 Limitations

(1) **Single split.** We use one leave-last-5-out partition by design; conclusions are read against seed spread, not split variance. No cross-validation is performed.

(2) **No significance testing.** Differences are compared against reported standard deviations; formal statistical tests are not run. Small deltas (e.g., VBPR vs MultiModalNCF HR@10) are explicitly left undecided.

(3) **Epoch budgets are not equalised.** MultiModalNCF runs 10 epochs against VBPR's 30 and BPR-MF/LightGCN's 50, so accuracy comparisons carry that asymmetry.

(4) **Training-time measurement conditions.** Hardware, warm/cold cache conditions are not recorded alongside reported seconds.

(5) **Second-domain slice represents heavy users only.** The Amazon slice is restricted to users with ≥ 20 ratings, chosen to match the MovieLens population rule, and does not describe the full Amazon population.

(6) **Content ablation effect is small.** Removing all content costs about one seed std on MovieLens and nothing measurable on Amazon.

(7) **Free-tier deployments sleep when idle.** Reported live behaviour may include cold-start latency on first request.

6 CONCLUSION

We presented CineRec, a practical comparative study of six recommendation models across two domains, with two published baselines and multi-seed validation. Our findings confirm that (1) model rankings are largely consistent across datasets, (2) content features provide measurable benefits in cold-start scenarios, and (3) a simple multi-modal extension of NeuMF can compete with established content-aware baselines like VBPR.

Future work includes: (1) evaluating on larger-scale datasets such as KuaiRec or Yelp, (2) exploring more sophisticated fusion strategies beyond simple concatenation, (3) conducting formal significance testing, and (4) extending the cold-start study to user cold-start as well as item cold-start. All code and evaluation scripts are available at: https://github.com/ElijahZhao/cinerec

ACKNOWLEDGMENTS

This project was developed as an independent study. The author thanks the open-source community for providing baseline implementations and datasets.

AI use disclosure: This project was assisted by AI tools throughout development and manuscript preparation, including code generation, debugging, and documentation writing. The author is solely

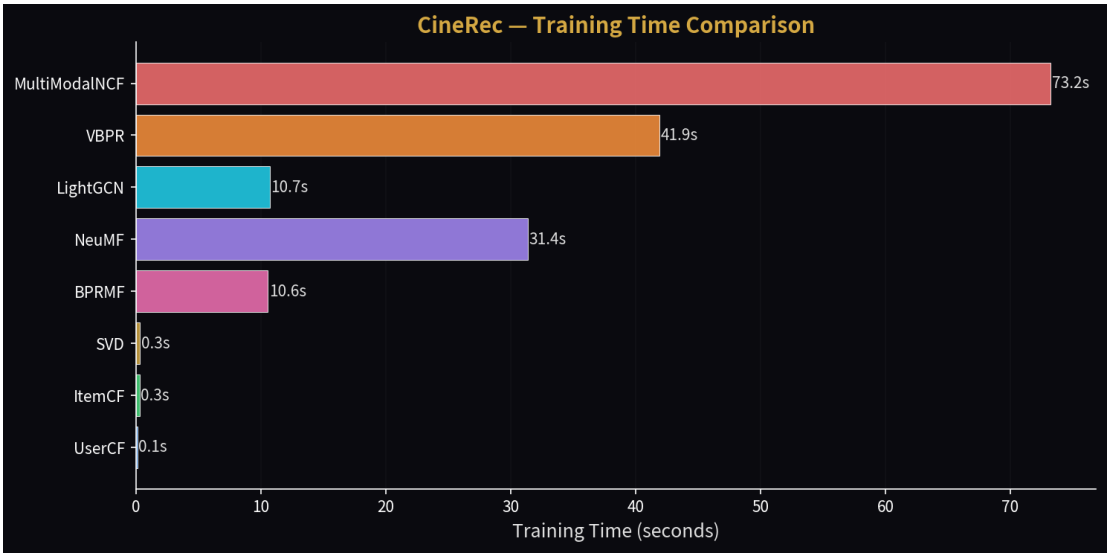


Figure 7: Training time per model (5-seed means, same runs as Table 2). Measurement conditions are not recorded; see the limitations.

responsible for the study design, the verification of all reported results, and the content of this paper.

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